



Md. Latifur Rahman Rafi

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About Me

Computer Science undergraduate and emerging researcher with interests in artificial intelligence, machine learning, and large language models (LLMs). Author of AI-focused research publications and actively engaged in academic research and technical communities. Experienced in mentoring students, conducting workshops, and promoting collaborative learning. Passionate about teaching, research, and supporting student success. Seeking a Teaching Assistant position to contribute to instruction, academic excellence, and AI research initiatives.

Relevant Courses Completed

- **Strong Performance:** Data Structures & Algorithms, Database Systems, Artificial Intelligence, Machine Learning
- **Completed:** Computer Networks, Computer Architecture, Object-Oriented Programming, Software Engineering

Teaching & Academic Experience

Teaching Assistant – Mastering DevOps, Ostad Limited Aug 2025 – Present

- Deliver one-on-one and live support for a DevOps course - resolving learner queries, monitoring progress, and providing timely feedback across sessions and community discussion channels
- Create and review quizzes, assignments, and live test questions, managing learner submissions to ensure timely, consistent evaluation
- Maintain a knowledge base of FAQs, learning materials, and pre-recorded videos covering Docker, CI/CD pipelines, cloud deployment (AWS), and infrastructure-as-code

Course Co-ordinator – AI Engineering MasterClass 2026 – Present

- Co-ordinating an ongoing course organized by Daffodil AI Club in collaboration with the Department of CSE, Daffodil International University
- Manage scheduling, instructor coordination, participant communication, and session facilitation

Education

Daffodil International University, B.S.C in Computer Science & Engineering Jan 2023 – Dec 2026

- CGPA: 3.06/4.00

Cantonment College, Cumilla Cantonment, Higher Secondary Certificate (HSC) 2018 – 2020

- GPA: 4.42/5.00

Publications

Calibrated and Explainable Credit Risk Assessment: Integrating SHAP Analysis with Feature Stability Validation 2026

- **Md. Latifur Rahman Rafi**, Ujjal Vowmick, Md. Rony, Prokrity Banarjee, Md. Minhajul Islam, Iftikher Ahmed Siam
- *2026 International Conference on Engineering and Frontier Technologies (ICEFronT2026, IEEE)* — **Accepted**
- Developed a unified credit risk framework integrating ROC-AUC, Brier Score, and SHAP with 5-fold validation (CV < 12%); LightGBM achieved 0.756 ROC-AUC with superior calibration.

- Md. Minhajul Islam, Md. Rony, **Md. Latifur Rahman Rafi**, Md Sabbir Ahmed, Md. Moshfiqur Rahman, Saima Sultana
- *2026 International Conference on Engineering and Frontier Technologies (ICEFronT2026, IEEE)* — **Accepted**
- Proposed a comparative ensemble ML framework (Random Forest, LightGBM, XGBoost) for clinical heart failure prediction; XGBoost achieved highest accuracy with balanced sensitivity and interpretability across clinical feature sets

Research Interests

- Machine Learning & Ensemble Methods
- Explainable AI (XAI) & Model Interpretability
- Large Language Models (LLMs) & Retrieval-Augmented Generation (RAG)
- Applied Artificial Intelligence & Backend Systems

Research Skills

- Machine Learning Modeling (ensemble methods, gradient boosting, classification pipelines)
- Calibration Analysis (Brier Score, reliability diagrams)
- Large Language Models - fine-tuning, Retrieval-Augmented Generation (RAG), context window management
- Literature Review & Academic Writing
- Experimental Design & Reproducible Research

Projects

Credit Risk Predict – Explainable Credit Scoring Pipeline

github.com/Credit_Risk_Predict

- Implementation behind the accepted ICEFronT 2026 (IEEE) publication - a unified credit-risk framework on the German Credit dataset integrating discrimination, calibration, and explainability
- Built the full pipeline: preprocessing, gradient-boosted modeling (LightGBM, ROC-AUC 0.756), 5-fold stability validation (CV < 12%), bias/fairness checks, and SHAP-based calibration and reliability plots
- **Tools:** Python, LightGBM, XGBoost, Scikit-learn, SHAP

ClusterMind AI – AI Scheduler for GPU Clusters

github.com/ClusterMind-AI

- Built for the DIU AI Innovation Hackathon 2026; real-time dashboard tracking per-node GPU, CPU, memory, temperature, and running jobs across a simulated cluster
- Trained an XGBoost scheduler that recommends optimal node placement for incoming jobs against live drifting metrics, designed to swap in real Prometheus/DCGM data with no code changes
- **Tools:** TypeScript, Python, XGBoost, React

RAG Pipeline – Retrieval-Augmented Generation from Scratch

github.com/RAG_Pipeline

- Built a modular RAG system component-by-component - custom embedder, vector store, retriever, and local LLM client (Ollama) - without relying on a framework wrapper
- Ingests CSV knowledge sources for context-grounded question answering over a configurable retrieval pipeline
- **Tools:** Python, Ollama, Vector Embeddings

E-Commerce ChatBot – Embeddings-Based Shopping Assistant

github.com/E-CommerceBot

- Developed a Django conversational assistant that answers product queries using an embeddings pipeline generated over the product catalog
- Implemented a model-manager layer for LLM interactions and a test harness for validating chatbot responses
- **Tools:** Python, Django, Embeddings, SQLite

SySToolkit – Unified OS Toolkit (Go + Rust)

github.com/SyStemToolKIT

- Built a unified command-line OS toolkit combining a concurrent TCP port scanner, virtual filesystem simulator, process monitor, and deadlock detection
- Designed a Go core CLI orchestrating feature modules, including a Rust-based AI-agent binary — demonstrating core OS concepts and multi-language systems programming
- **Tools:** Go, Rust

Vote-Ai – Full-Stack Online Voting Platform

github.com/Vote-Ai

- Built a full-stack voting platform with Google OAuth authentication and MySQL, supporting club-based voting with real data population and admin tooling
- Documented the system with a complete UML suite - use-case, class, sequence, activity, and architecture diagrams
- **Tools:** Python, Django, Google OAuth, MySQL

Student Portal – DIU Student Portal (Go + React)

github.com/Student-Portal

- Developed a Go backend (routes, handlers, models) serving student academic data, paired with a modern React + Tailwind frontend
- Built a responsive dashboard with financial summaries and academic-performance charts using Chart.js
- **Tools:** Go, React, Tailwind CSS, Chart.js

Todo CI/CD – Automated Deployment Pipeline

github.com/todo_ci-cd

- Instrumented a Django task application with GitHub Actions workflows for automated testing and continuous deployment
- Demonstrates a working CI/CD pipeline from commit to deploy, backing the DevOps skills listed on the CV
- **Tools:** Python, Django, GitHub Actions

Technical Skills

Languages: Python, Go, Rust, C

Frameworks & Libraries: Django, React, Tailwind CSS

Machine Learning & AI: Scikit-learn, XGBoost, LightGBM, SHAP, RAG, Vector Embeddings, Ollama

Databases: MySQL, SQLite

DevOps & Tools: Git, GitHub Actions, Docker, NGINX, Netdata, Terraform, AWS EC2, Google OAuth

Operating Systems: Linux (Arch-based), Ubuntu Server

Leadership & Extracurricular Activities

Founding Organizing Secretary, CSE AI Centre & Daffodil AI Club (2025 – 2026)

🌟 Certifications

Planned and executed the following events as organizer:

- **ModalX-AI Challenge** – AI competition for students to solve real-world problems using machine learning and AI tools
- **Daffodil AI Club Official Inauguration Ceremony** – Coordinated logistics and program for the club's founding event
- **Webinar: AI Engineering MasterClass** – Managed speaker coordination, participant registration, and online delivery
- **Seminar: Digital Twin – Transforming the Future of Intelligent Systems** – Technical seminar on Digital Twin technology and intelligent systems
- **Online Session: Data to Dashboard – Hands-on Power BI Session** – Practical live session on data visualization with Power BI
- **Seminar: How to Choose a Research Topic & Start Research** – Academic guidance seminar for undergraduate students beginning their research journey

- **Seminar: What Does the Industry Really Expect from Future AI & Machine Learning Engineers?** – Industry-focused seminar on the real-world skills, tools, and expectations for aspiring AI and ML engineers entering the workforce
- **ML Bootcamp for Beginners** (Powered by Shohoj Coding) – Coordinated a hands-on machine learning bootcamp for beginners as Club Coordinator, in collaboration with Shohoj Coding

References

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